

		SET 1	Total Pages: 4
Reg No.: _____			Name: _____
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY FIRST SEMESTER (R) M Tech DEGREE EXAMINATION, JANUARY 2025			
Discipline: INTERDISCIPLINARY			
Course Code : 221RGE100			
Course Name : RESEARCH METHODOLOGY AND IPR			
Max. Marks: 60			Duration: 2.5 Hours
PART A			
		Answer all questions. Each question carries 5 marks	Marks
	1	Construct a list of characteristics of good research problem and explain	(5)
	2	What are the different methods of research gap analysis	(5)
	3	Apply the concept of hypothesis in a research scenario. Explain how formulating a clear hypothesis can guide the research process and impact the outcomes of your study	(5)
	4	Enlist the guidelines for preparation of good presentation slides.	(5)
	5	Illustrate with necessary diagram the different stages from filing to grant of a patent.	(5)
	6	What are the difference stages in the realization of a paper.	(5)
PART B			
Read the given article and write a report that addresses the following issues			
	7	Optical Flow Sensor and Laser Sensor-Based Automatic Target Following Robot Based on optical flow sensor and laser sensor technology, an automatic following robot is designed by this project. L-K optical flow method optimized by Gaussian-Laplace operator and laser algorithm to obtain the coordinates and speed of the following target, and the Kalman filter is used to remove noise. The simulation results show that the robot has good robustness and adaptability by processing the adaptive weighting algorithm and fuzzy algorithm on the data. With the expansion of the application field of intelligent robots, tremendous achievements have been made in the intelligent economy and intelligent society.	

	People expect that intelligent robots can replace humans in more fields to complete complex tasks. The rise of smart devices such as DJI drones and Xiaomi home assistants on the Chinese market shows that service robots are becoming a new research trend. Therefore, the automatic following robot which can play a role in various fields is designed by this paper. In the field of automatic following robots, most researches use a single sensor to locate the following target, but sensors with high measurement accuracy and large range can hardly avoid the problem of the high cost. Due to the limitation of the sensor, the low-cost sensor cannot provide high-precision and large-scale data at the same time. Therefore, this paper uses laser and optical flow dual sensors to simulate the research of automatic target tracking, which has high practical value. In order to achieve tracking of the target, three laser sensors and one optical flow sensor are installed on the robot. The laser sensor is used to obtain the accurate distance and speed of the following target in the y-axis direction, and the optical flow sensor is used to obtain the rough distance and accurate speed of the following target in the x-axis direction, which is determined by the weighted fusion of the information returned by the two sensors final coordinate and speed information. Then, the robots main control chip controls the motion mechanism to make corresponding responses through coordinate and speed information and realizes automatic tracking of the target. Schematic of automatically following Among them, the main control chip of the robot is the RT1021 microcontroller. The software of the one-chip computer includes the Gaussian Laplace pyramid algorithm and the L-K optical flow method to obtain more accurate optical flow information. Through the optimization algorithm and fuzzy algorithm processing, the robustness of the robot can be enhanced.	
--	---	--

		In order to test the effectiveness of the research method, a given error correction experiment is conducted through Matlab, a fuzzy controller is designed with fuzzy rules, and then the fusion information obtained by the adaptive weighting algorithm is input to the fuzzy controller. In this way, the error is corrected by the fuzzy controller. The simulation results show that the adaptive weighting algorithm and fuzzy PID-controlled tracking system have good self-tuning parameters and strong adaptability, and can quickly and accurately achieve automatic tracking of the target. The single-sensor following robot is limited in terms of information acquisition and processing capabilities. For complex tasks and changeable working environments, the ability of the single-sensor following robot is even more insufficient. The following robot based on optical flow sensor and laser sensor for data fusion processing has stronger advantages than the single-sensor following robot. The automatic following robot based on optical flow sensor and laser sensor can better respond to the changeable environment and complete the corresponding follow-up tasks in the complex environment. In order to test the effectiveness of the robot and reduce the cost of development, this paper simulates the corresponding algorithm through Matlab. Experimental results show that the system can effectively perform data fusion and complete corresponding follow-up tasks through fuzzy algorithms. The system is stable and reliable, and provides a new idea for the study of multi-sensor automatic following robots.	
	a	What is the main research problem addressed?	(4)
	b	Identify the type of research	(4)
	c	Discuss the limitations of this research if any?	(4)
	d	Discuss the significance of the study	(6)

	e	Discuss appropriateness of the methodology used for the study	(6)
	f	Analyse and summarize the simulation results and contributions by the authors	(6)
